

Curriculum Vitae — Dongkuan Zhang

Personal Information

Name	Dongkuan Zhang	
Nationality	Chinese	
Date of Birth	02.05.1995	
Location	Dalian, Liaoning, China	
Phone	+86 15510689505	
Email	dkz@mail.dlut.edu.cn	
Languages	Mandarin (Native), English (Fluent)	

Education

2023 – Now	Dalian University of Technology Ph.D. in Environmental Science and Engineering (expected 2027). Supervisor: Prof. Guozhao Ji
2020 – 2023	Minzu University of China M.S. in Environmental Science and Engineering
2013 – 2017	Liaoning Petrochemical University B.S. in Petroleum Engineering

Work Experience

2017 – 2019	Zhejiang Petrochemical Co., Ltd. Diesel hydrocracking unit operator — equipment monitoring, process adjustment, sample collection and analysis, troubleshooting, safety management and record keeping.
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Joint Training & Visiting

2025 – 2026	Cross Laboratory, Institute of Science Tokyo Joint PhD student supervised by Prof. Jeffrey S. Cross. Funded by the China Scholarship Council (CSC).
2026.09 – 2027.03	University of Luxembourg Joint PhD research visit (6 months), supervised by Prof. Bradley P. Ladewig. Funded by the DUT Global Vision Project.

Research Interests

- Intelligent multiphase flow modeling for waste incineration:** combining physics-based CFD with AI-driven flow field prediction.

- Eulerian-Lagrangian modeling of multiphase flow, combustion and heat transfer (lab to industrial scale).
- Machine-learning and neural-network surrogates (e.g. input-convex neural networks) for fast flow prediction and non-Newtonian fluid modeling.
- Flow field design, anti-clogging and operational optimization of incinerators, circulating fluidized beds and twin-fluid nozzles.

Academic Output Snapshot

Total citations **60** · h-index **4** · i10-index **1** (updated 2026-09-01)

- Journal articles: **11** papers (incl. first/co-first author 5 papers)
- Patents: **9** patents (granted invention patents, utility models and published applications)
- Software copyrights: **2** items
- Group standards: **1** (participating drafter)
- Technology transfer: **1** patent transfer, RMB 400,000

Journal Articles ⁽¹¹⁾

2026	Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow D Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G Ji <i>Chemical Engineering Research and Design</i> · link
2026	Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions L Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G Ji <i>Processes</i> 14 (13), 2189 · link
2026	Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ... <i>Industrial & Engineering Chemistry Research</i> · link
2026	Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ... <i>Separation and Purification Technology</i> , 137736 · link · 2 citations
2025	Input-convex neural network modeling of rheological properties of non-Newtonian fluids in chemical distillation residues Dongkuan Zhang, Zhiqiang Chu, Hong Tian, Guozhao Ji <i>Journal of Xi'an Jiaotong University</i> , 2025, 59(12): 80-91 · link

2025	The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator
	J Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G Ji
	<i>Processes</i> 13 (12), 3928 · link
2025	Eulerian-Lagrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds
	J Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G Ji
	<i>Applied Thermal Engineering</i> , 127937 · link · 5 citations
2025	Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration
	D Zhang, T Anjum, Z Chu, JS Cross, G Ji
	<i>Renewable and Sustainable Energy Reviews</i> 221, 115895 · link · 39 citations
2023	Experimental study on sediment disturbance during deep-sea polymetallic nodule collection
	Dongkuan Zhang, Meilin Liu, Jianxin Xia
	<i>Mining and Metallurgical Engineering</i> , 2023, 43(3): 20-23 · link · 4 citations
2022	Flow field parameter matching of deep-sea polymetallic nodule fluidization acquisition head
	Dongkuan Zhang, Lei Guan, Huade Cao, et al.
	<i>Journal of Ocean Technology</i> , 2022, 41(4): 53-60 · link · 4 citations
2021	Comparison and analysis of deep-sea polymetallic nodule collection methods and structural parameters
	Lei Guan, Dongkuan Zhang, Yuchao Xia, Jianxin Xia
	<i>Journal of Ocean Technology</i> , 2021, 40(5): 62-70 · link · 3 citations

See the full publication list with citation counts

Patents ⁽⁹⁾

2026	A dual-spray-gun self-cleaning spray system
	Guozhao Ji, Dongkuan Zhang, Junchao Yang, Jiacheng Dai, Hong Tian, Tong Ren
	<i>China Invention Patent Application Publication No. CN121972345A (Appl. No. CN202610181384.4, published May 5, 2026)</i>

2026	An external-mixing dual-fluid atomizing spray gun Guozhao Ji, Dongkuan Zhang <i>China Invention Patent Application Publication No. CN121847357A (Appl. No. CN202610181380.6, published Apr 14, 2026)</i>
2025	A device for self-settling fly ash in a waste incinerator Guozhao Ji, Dongkuan Zhang <i>China Utility Model Patent No. ZL 2024 2 2052228.7 (Granted CN 223137888 U)</i>
2024	A device, method and application for self-settling fly ash in a waste incinerator Guozhao Ji, Dongkuan Zhang <i>China Invention Patent Application Publication No. CN118912512A (Appl. No. CN202411163901.2, published Nov 8, 2024)</i>
2024	A fluidized bed reactor based on Tesla Valve Guozhao Ji, Dongkuan Zhang <i>China Invention Patent ZL202311155946.0 (granted; publication No. CN117504737B, Oct 1, 2024) · Scholar</i>
2023	A device, method and application for self-settling fly ash in a waste incinerator Guozhao Ji, Dongkuan Zhang <i>China Invention Patent Application Publication No. CN117308105A (Appl. No. CN202311250560.8, published Dec 29, 2023)</i>
2022	Parameter matching method for polymetallic nodule collection head Jianxin Xia, Lei Guan, Fangfang Luan, Huade Cao, Dingbang Wei, Dongkuan Zhang <i>China Invention Patent Application Publication No. CN114329824A (Appl. No. CN202111566075.2) · Scholar</i>
2022	A design method for flow field of polymetallic nodules collector Jianxin Xia, Dongkuan Zhang, Fangfang Luan, Huade Cao, Dingbang Wei, Lei Guan <i>China Invention Patent ZL202111562069.X (granted); application publication No. CN113946927A · Scholar</i>
2016	A probe for petroleum exploration Chengchen Wu, Boyi Zhou, Shuo Ma, et al. <i>China Utility Model Patent No. CN201620273762.3 (granted, Aug 3, 2016) · Scholar</i>

Software Copyrights

2025-09 Dual Spray Gun Self-Cleaning Control System (DSGSCS) V1.0

China Computer Software Copyright Registration

2025-07 Nozzle (Pipe) Blockage Risk Monitoring Software (NRMS) V1.0

China Computer Software Copyright Registration

Standards

2025 Technical Requirements for Hydrogen Production from Waste Based on High-Temperature Pyrolysis and Gasification

Dongkuan Zhang (participating drafter), et al.

Group/Industry Standard (participating author)

Research Projects & Technology Transfer

2023 – 2024 Self-settling fly ash technology for waste incinerators (technology transfer)

As one of the completing researchers, transferred self-developed patent achievements and related research results to industry (RMB 400,000). Personally awarded RMB 20,000 cash incentive by the university.

Honors & Awards

2026.06

Best Poster Award, The First International Symposium on Environmental Nanotechnology, Dalian, China.

Poster: *Optimizing hazardous waste incineration in a full-scale rotary kiln via co-firing and staged air supply*
· certificate

2026.04

DUT Global Vision Project Funding (University of Luxembourg, 6-month joint PhD visit).

2025.12

2025 Chunhui Overseas Young Excellent Achievement Collection (MOE of China). Project: "Crisis-Flow Control: Flow-field-based Hazardous Waste Incineration & Disposal Technology and Equipment."

2025.09

National Scholarship for Doctoral Students (Ministry of Education of China).

2025.11

12th Japan-China-Korea Joint Symposium on Clean Energy and Sustainable Environment, Yamanakako, Japan — Presentation Certificate.

2025.01

Drafter Certificate, Group Standard T/CITS 242-2025, China Inspection and Testing Society.

2024.12

"Environmental Engineering Star" (Dalian University of Technology).

2024.11

Award for Excellent Presentation, 2024 Korea-China-Japan Joint Symposium on Energy and Environment (Yonsei University).

2024.11

China Scholarship Council (CSC) Government-Sponsored Studentship (CSC No. 202406060182).

2024

Cash incentive for technology transfer (RMB 20,000), Dalian University of Technology.

Service & Professional Activities

- Drafter of group standard T/CITS 242-2025, China Inspection and Testing Society.
- Peer reviewer for Chemical Engineering Research and Design, Separation and Purification Technology, Applied Thermal Engineering, Journal of Cleaner Production, and related journals.

Technical Skills

CFD	ANSYS Fluent, Star-CCM+, COMSOL Multiphysics; multiphase flow, combustion and heat transfer modeling.
Programming	Python (PyTorch, NumPy, SciPy), MATLAB; data analysis and visualization.
Experiment	Flow-field testing, heat-transfer experiments, characterization.
